

Oscillations

Periodic Motion:- Any motion that repeats itself in equal intervals of time.

Oscillatory Motion:- If a particle in periodic motion moves back and forth over the same path.

Harmonic Motion:- The displacement of a particle in periodic motion can always be expressed in terms of sine and cosine functions therefore oscillatory motions are called Harmonic motions.

Oscillations

- The world is full of oscillatory motions
 - A child on a swing
 - A mass attached to a spring
 - A guitar string being played
 - Swinging pendulum
 - Atoms in molecules or in solid lattice
 - Air molecules as a sound wave passes by
 - Radio waves, microwaves and visible light are oscillating magnetic and electric field vectors

Simple Harmonic Motion

- Simple harmonic motion (SHM) refers to a certain kind of oscillatory, or wave-like motion that describes the behavior of many physical phenomena:
 - a pendulum
 - a bob attached to a spring
 - low amplitude waves in air (sound), water, the ground
 - the electromagnetic field of laser light
 - vibration of a plucked guitar string
 - the electric current of most AC power supplies

OSCILLATIONS & SIMPLE HARMONIC MOTION

1) Oscillatory Motion

Oscillatory motion is the motion in which a body moves to and fro about a fixed mean position.

Examples:

- Motion of a simple pendulum
- Motion of a mass attached to a spring
- Vibrations of a tuning fork

2) Periodic Motion

A motion which repeats itself after equal intervals of time is called periodic motion.

Time Period (T): Time taken to complete one full oscillation.

Frequency (f): Number of oscillations per second.

Relation between frequency and time period:

$$f = \frac{1}{T}$$

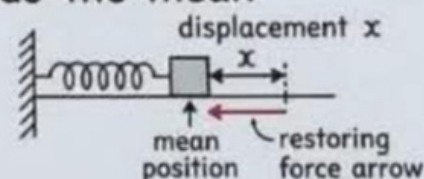
SI unit of frequency: Hertz (Hz)

3) Simple Harmonic Motion (SHM)

Simple harmonic motion is a type of oscillatory motion in which restoring force is directly proportional to displacement from the mean position and always directed towards the mean position.

Mathematical condition for SHM: $F \propto -x$

Therefore: $F = -kx$



Where: k = force constant, x = displacement from mean position

4) Acceleration in SHM

Acceleration is proportional to displacement and directed towards the mean position.

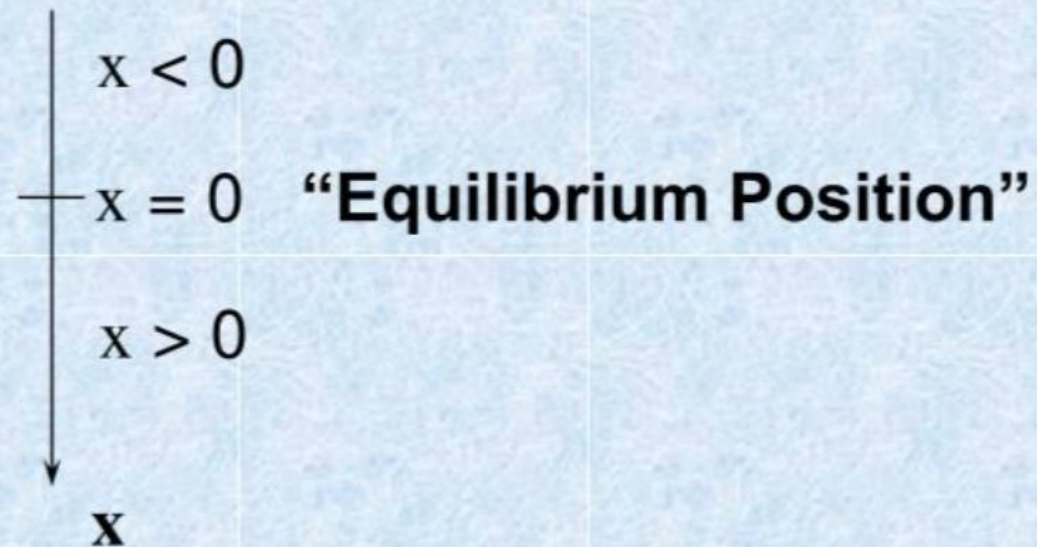
$$a = -\omega^2 x$$

Important Points:

- Restoring force always acts toward the mean position
- SHM is a special type of oscillatory motion
- Motion is symmetric about the mean position

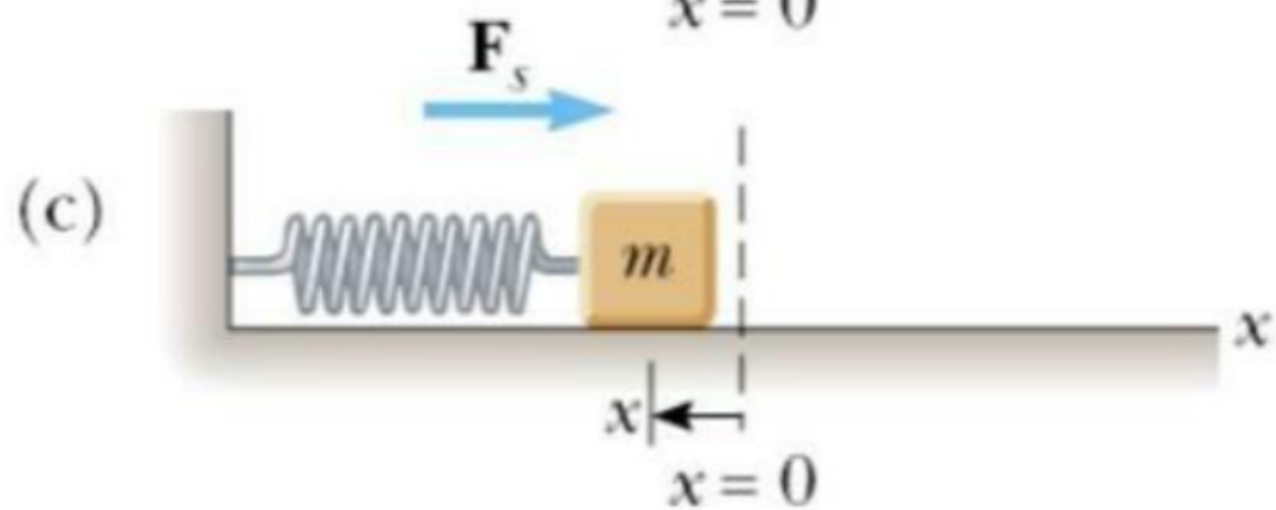
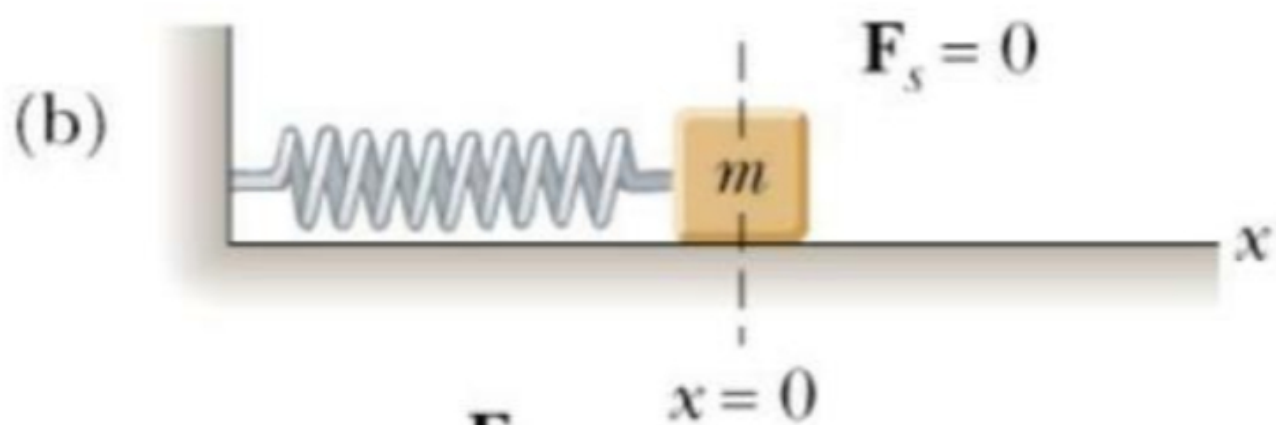
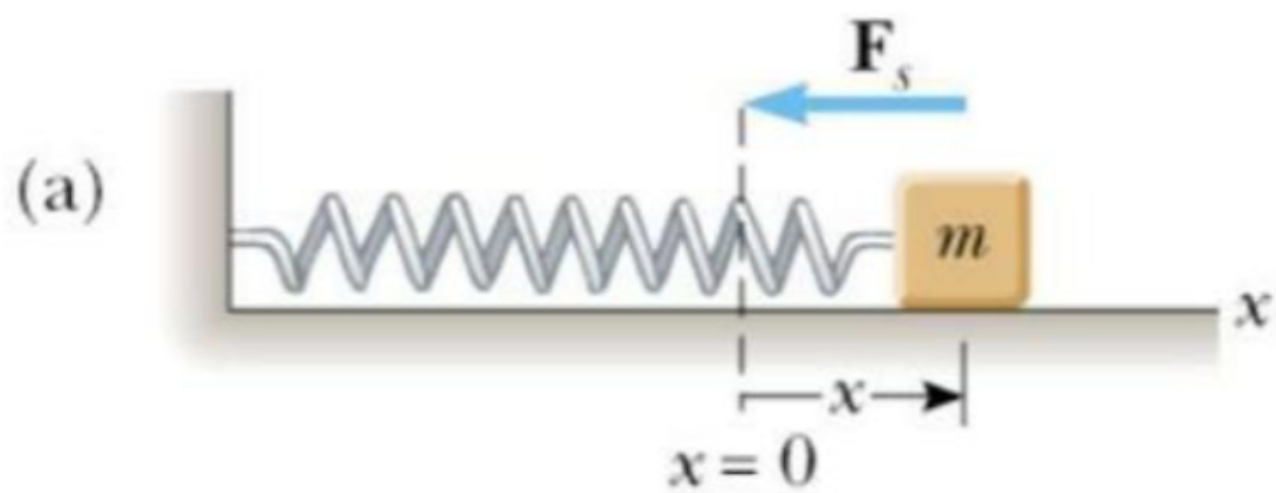
Simple Harmonic Motion

❖ Mass Attached to a Spring



Simple Harmonic Motion

- ❖ Mass Attached to a Spring
 - When the mass is stationary, it is in its equilibrium position (Net force = 0).
 - If we pull the mass down slightly and then release it, what happens?
 - Graph the displacement from equilibrium as a function of time.
 - $x(t) = A \cos(\omega t)$
 - where A = amplitude = a constant, and
 - ω = angular frequency = a constant.
 - We'll learn more about these quantities soon.



Simple Harmonic Motion

❖ Equations for this *Special Case*

– **Displacement** from Equilibrium:

$$x(t) = A \cos (\omega t)$$

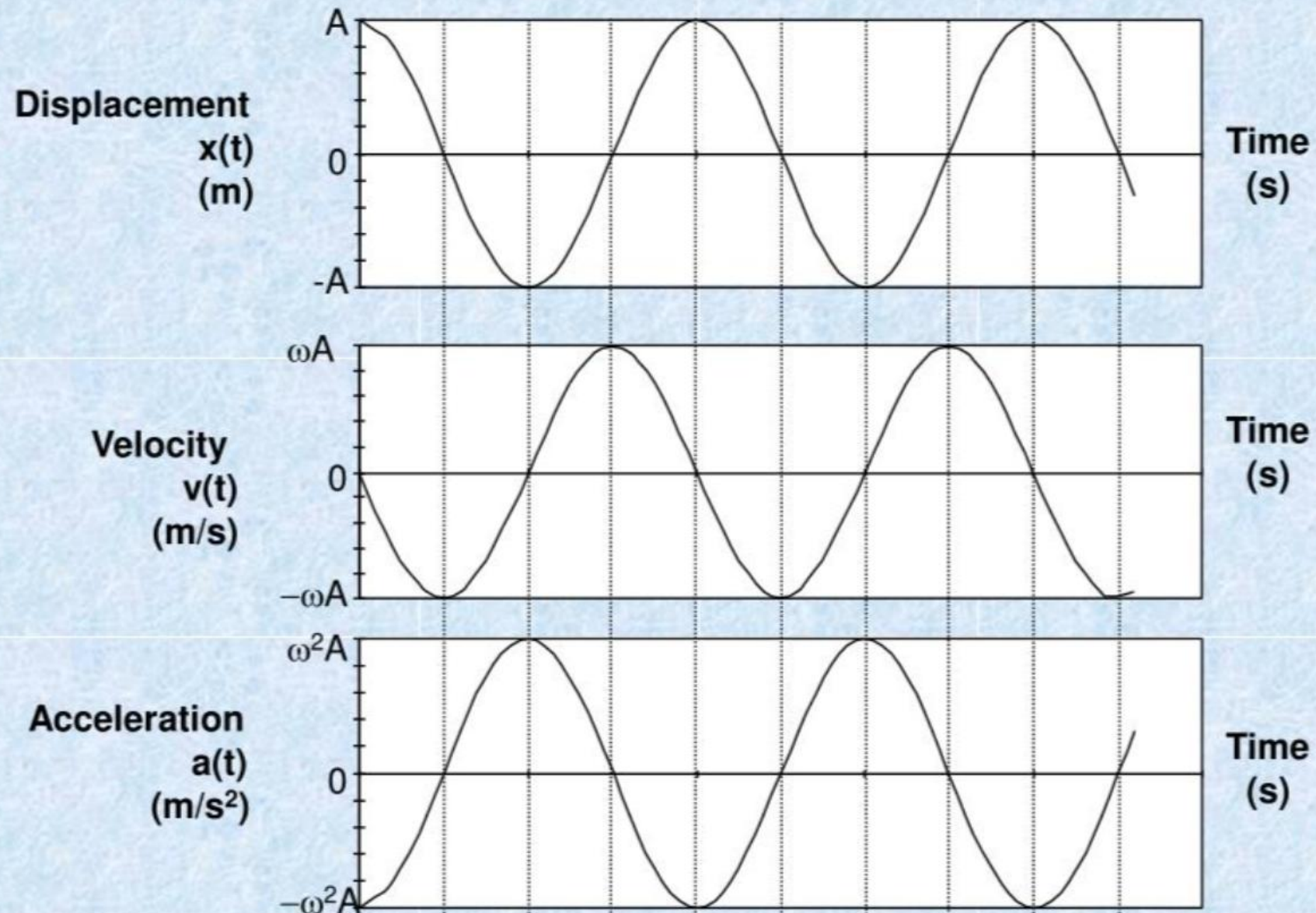
– **Velocity:**

$$v(t) = \frac{dx}{dt} = -\omega A \sin (\omega t)$$

– **Acceleration:**

$$a(t) = \frac{dv}{dt} = -\omega^2 A \cos (\omega t)$$

Simple Harmonic Motion



Simple Harmonic Motion

❖ Definitions of Terms

- **Amplitude = A** = the maximum displacement of the moving object from its equilibrium position.
 - (unit = m)
- **Time Period = T** = the time it takes the object to complete **one full cycle** of motion.
 - (unit = s)
- **Frequency = f** = the number of cycles or vibrations per unit of time.
 - (unit = cycles/s = 1/s = Hz = hertz)

Simple Harmonic Motion

❖ Definitions of Terms (continued)

– **Angular Frequency = ω**

(unit = radians/s = rad/s)

$$\omega = 2\pi f = \frac{2\pi}{T}$$

– **Phase Constant = Phase Angle = ϕ**

□ (unit = radians)

■ In general, simple harmonic motion cannot be described by a “pure” sine or cosine function, so a phase constant, ϕ , or phase angle must be introduced.

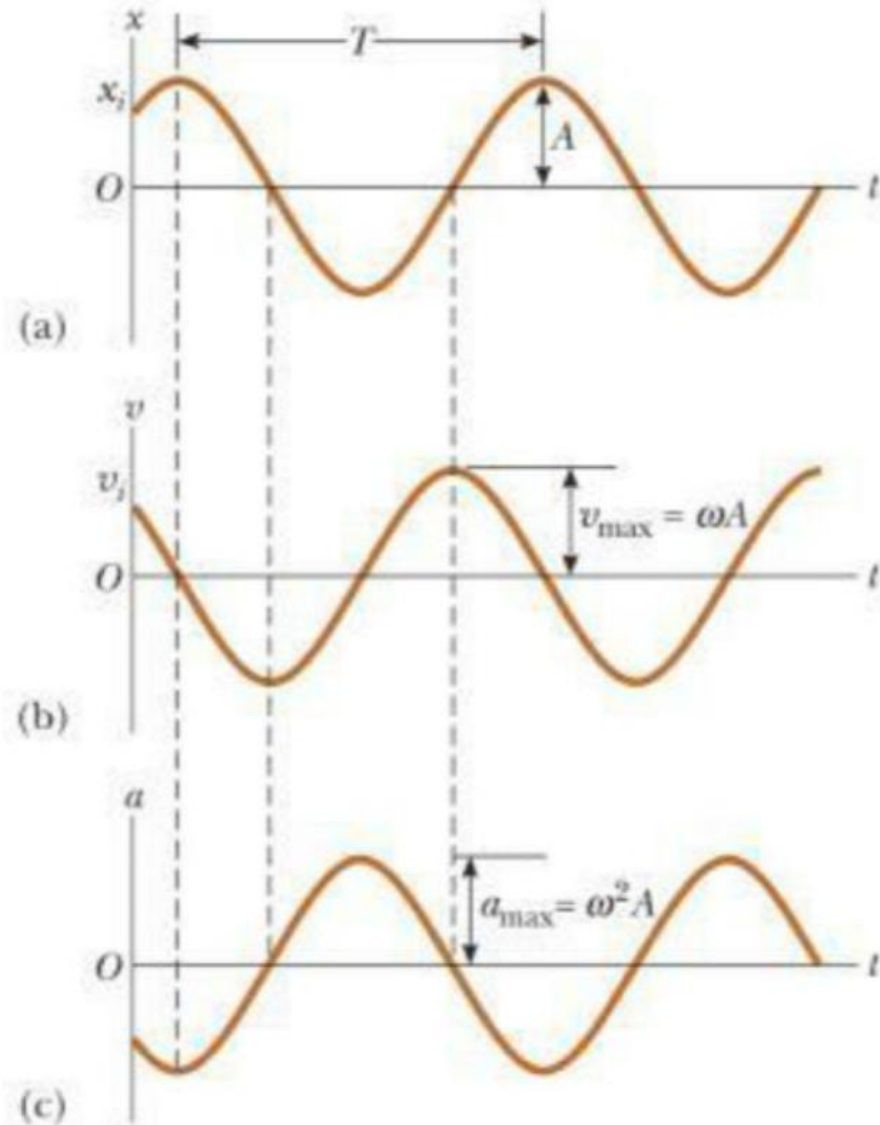
■ E.g., **$x(t) = A \cos (\omega t + \phi)$**

■ $(\omega t + \phi)$ is called the phase of the motion

Summary Graphs for SHM – *General Case*

In the most general case, the displacement graph doesn't begin at a peak.

This means that $\phi \neq 0$.



Simple Harmonic Motion

❖ General Equations

– **Displacement** from Equilibrium:

$$\mathbf{x(t) = A \cos (\omega t + \phi)}$$

– **Velocity:**

$$\mathbf{v(t) = \frac{dx}{dt} = -\omega A \sin (\omega t + \phi)}$$

– **Acceleration:**

$$\mathbf{a(t) = \frac{dv}{dt} = -\omega^2 A \cos (\omega t + \phi)}$$

TIME PERIOD OF SIMPLE HARMONIC MOTION

1) Time Period (T)

The time taken by a particle to complete one full oscillation is called the time period.

Relation with frequency:

$$T = \frac{1}{f}$$

Where:

T = time period

f = frequency

2) Angular Frequency

Angular frequency is related to time period.

$$\omega = \frac{2\pi}{T}$$

$$\omega = 2\pi f$$

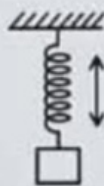
3) SHM of Mass-Spring System

For a mass attached to a spring:

Restoring force: $F = -kx$

Time period of spring system:

$$T = 2\pi \sqrt{\frac{m}{k}}$$



Where:

m = mass attached to spring

k = spring constant

4) SHM of Simple Pendulum

For small oscillations, the motion of a simple pendulum is SHM.

Time period of simple pendulum:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

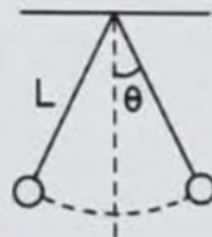
Where:

L = length of pendulum

g = acceleration due to gravity

5) Important Points

- Time period of simple pendulum does not depend on mass
- Time period does not depend on amplitude (for small oscillations)
- Time period increases if length increases
- Time period decreases if gravity increases



Equilibrium Position

- a point where the acceleration of the body undergoing oscillation is zero.
- At this point, the force exerted on the body is also zero.

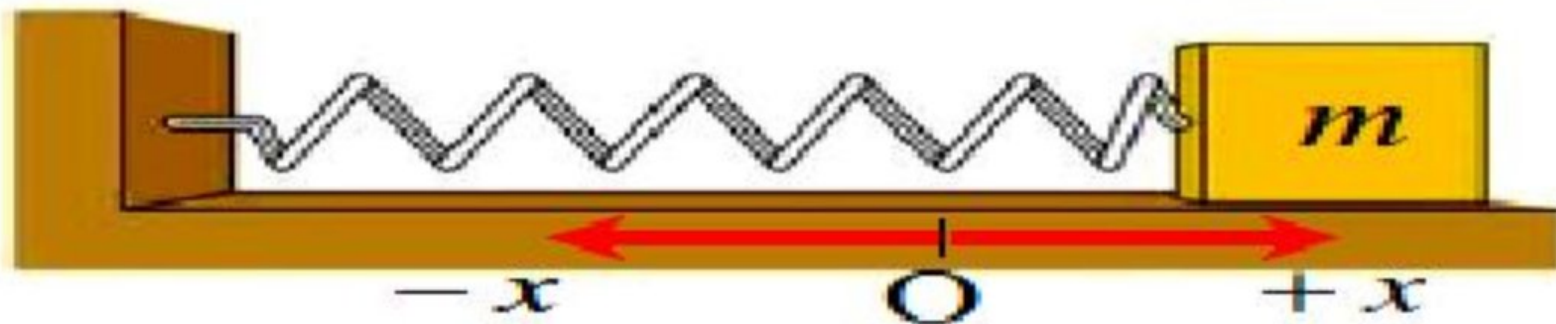
Restoring Force

- the force which causes simple harmonic motion to occur. This force is proportional to the displacement from equilibrium & always directed towards equilibrium.

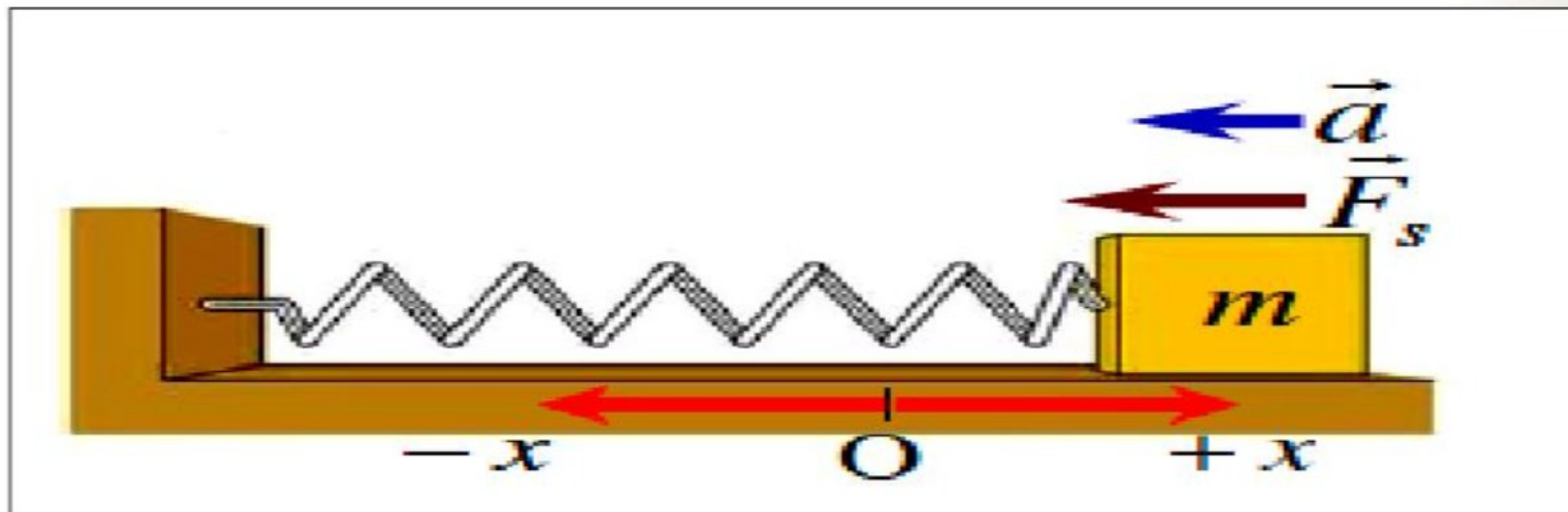
$$F_s = -k x$$

Equation for SHM

- Consider a system that consists of a block of mass, m attached to the end of a spring with the block free to move on a horizontal, frictionless surface.



- when the block is displaced to one side of its equilibrium position & released, it moves back & forth repeatedly about a maximum values of displacement x .
- Maximum value of x is called amplitude, A
- It can be negative ($-$) or positive ($+$).



- the spring exerts a force that tends to restore the spring to its equilibrium position.
- Given by Hooke's law :

$$F_s = -k x$$

$$F_s = -k x$$

- F_s is known as restoring force.
- Applying Newton's 2nd Law to the motion of the block :

$$F_{net} = ma$$

$$-k x = m a$$

$$a = -\frac{k}{m} x \quad \left(\frac{k}{m} : \text{constant value} \right)$$

- denote ratio k/m with symbol ω^2 :

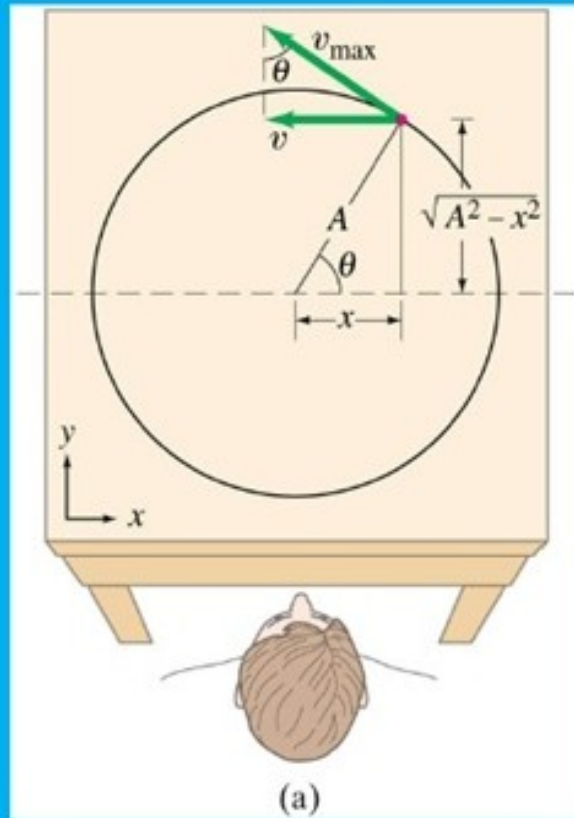
$$a = -\omega^2 x \quad [\text{equation for SHM}]$$

- any system that satisfy this equation is said to exhibit Simple Harmonic Motion (SHM)
- from above equation, we find that:

$$a \propto -x$$

- the acceleration, a is proportional to the displacement of the block & its direction is opposite the direction of the displacement.

SHM & Uniform circular motion



Let us consider the uniform circular motion of a particle.

The motion of the projection on x axis or y axis will be SHM

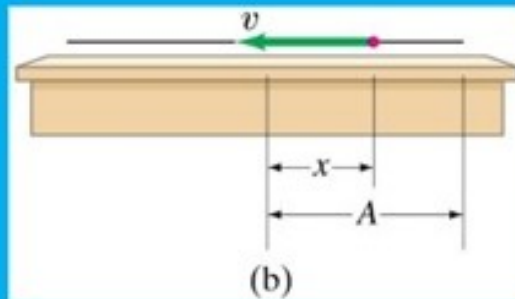
$A \Rightarrow$ radius of the circular path

$\theta \Rightarrow$ Angular displacement

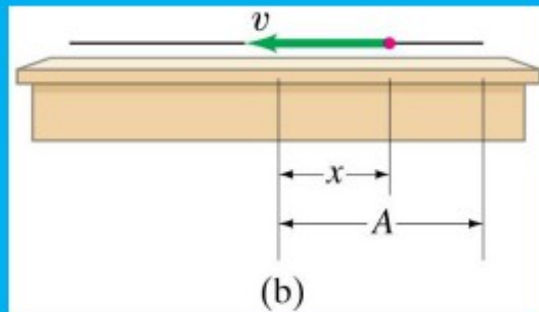
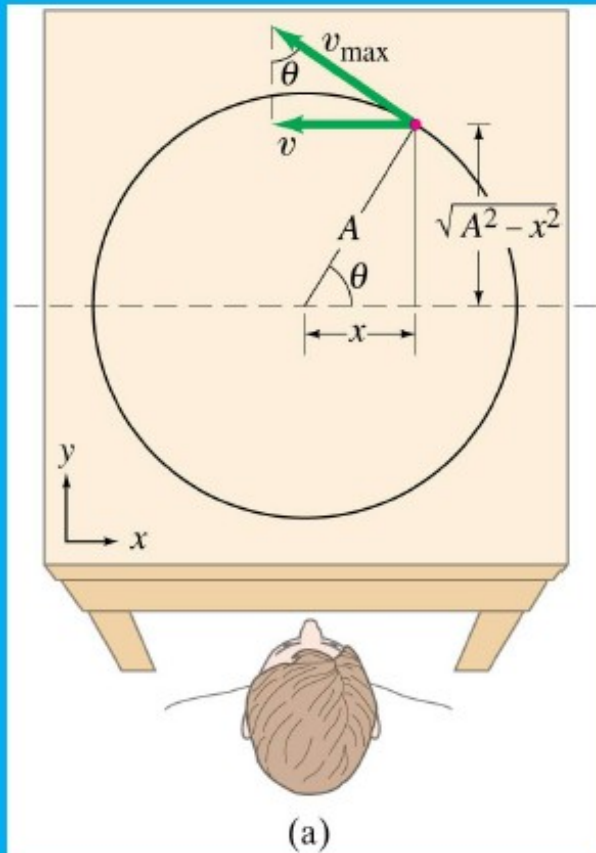
Then the displacement of the projection on x axis (or y axis) from the center (the mean position) of the circle is

$$x = A \cos (\theta) , y = A \sin (\theta)$$

Here $A \Rightarrow$ Amplitude of oscillation.



SHM & Uniform circular motion



If $\omega \Rightarrow$ uniform angular velocity
 $\Rightarrow \theta = \omega t$

Hence $x = A \cos \omega t$

Velocity $v = \frac{dx}{dt} = -A\omega \sin \omega t = -A\omega \sin \theta$

Acceleration $a = \frac{dv}{dt} = -A\omega^2 \cos \omega t = -\omega^2 x$
 $\Rightarrow a \propto -x$

So, motion of the projection is SHM

Energy of S.H.M

Potential Energy: Force $F = -kx$

For small displacement dx , work done

$$dw = -Fdx = -(-kx)dx = +kxdx$$

Hence total work done for displacement x ,

$$W = \int_0^x dw = \int_0^x kxdx = \frac{1}{2}kx^2$$

This work done will be store as P.E in the oscillator

$$P.E = U = \frac{1}{2}kx^2$$

$$\text{When } x = A, U = U_{\max} = \frac{1}{2}kA^2$$

Hence Total Energy = $E = K + U$

$$\Rightarrow E = \frac{1}{2}k(A^2 - x^2) + \frac{1}{2}kx^2$$

$$\Rightarrow E = \frac{1}{2}kA^2 = K_{\max} = U_{\max}$$

Energy of SHM

Total energy of SHM = Kinetic Energy (K.E) + Potential Energy (P.E)

Kinetic Energy: $K = \frac{1}{2}mv^2$

$$x = A \cos \omega t$$

$$\Rightarrow v = \frac{dx}{dt} = -A\omega \sin \omega t = -A\omega \sin \theta$$

$$\text{Hence K.E} = \mathbf{K} = \frac{1}{2}\mathbf{mv}^2 = \frac{1}{2}\mathbf{m}(-A\omega \sin \omega t)^2$$

$$\Rightarrow \mathbf{K} = \frac{1}{2}\mathbf{mA}^2\omega^2\sin^2\omega t$$

$$\Rightarrow K = \frac{1}{2}m\omega^2 A^2(1 - \cos^2 \omega t)$$

$$\omega^2 = K/m$$

$$\Rightarrow K = \frac{1}{2}m\omega^2(A^2 - x^2)$$

$$K = m\omega^2$$

$$\Rightarrow K = \frac{1}{2}k(A^2 - x^2)$$

$$\text{When } x = 0, K = K_{\max} = \frac{1}{2}kA^2$$

Total Mechanical Energy

Total energy in SHM is constant.

Total Energy = K.E. + P.E.

$$E = \frac{1}{2} m \omega^2 A^2$$

Important Points:

- Total energy remains constant during motion
- Energy continuously changes between kinetic and potential
- At mean position $\rightarrow$ energy is fully kinetic
- At extreme position $\rightarrow$ energy is fully potential

Energy Variation

At Mean Position ($x = 0$):

K.E. = maximum

P.E. = 0

At Extreme Position ($x = A$):

K.E. = 0

P.E. = maximum

